



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)
Machine Learning

Lecture 11: Support Vector Machines

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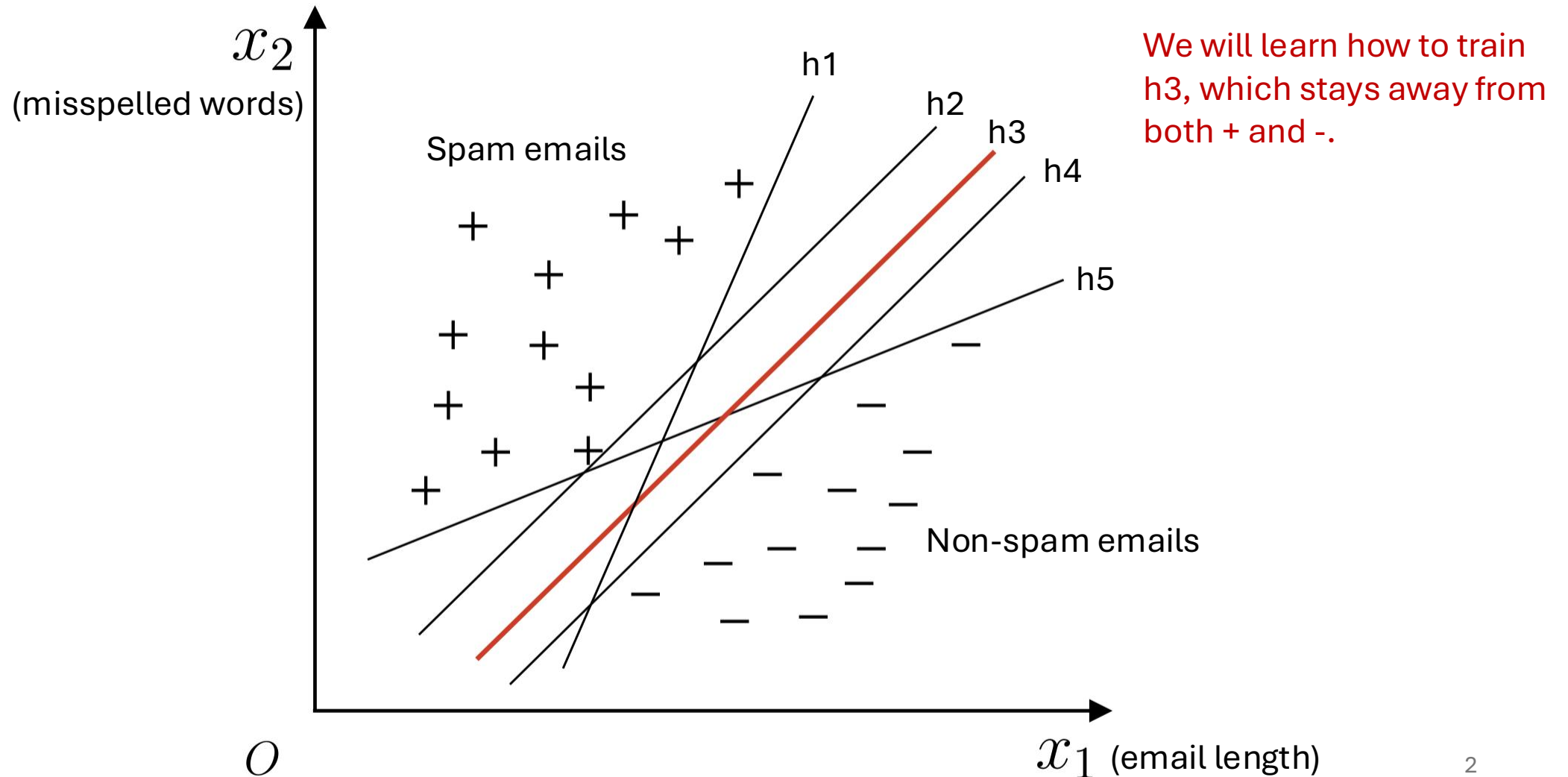
Department of Computer Science

Sep 29, 2026

Today

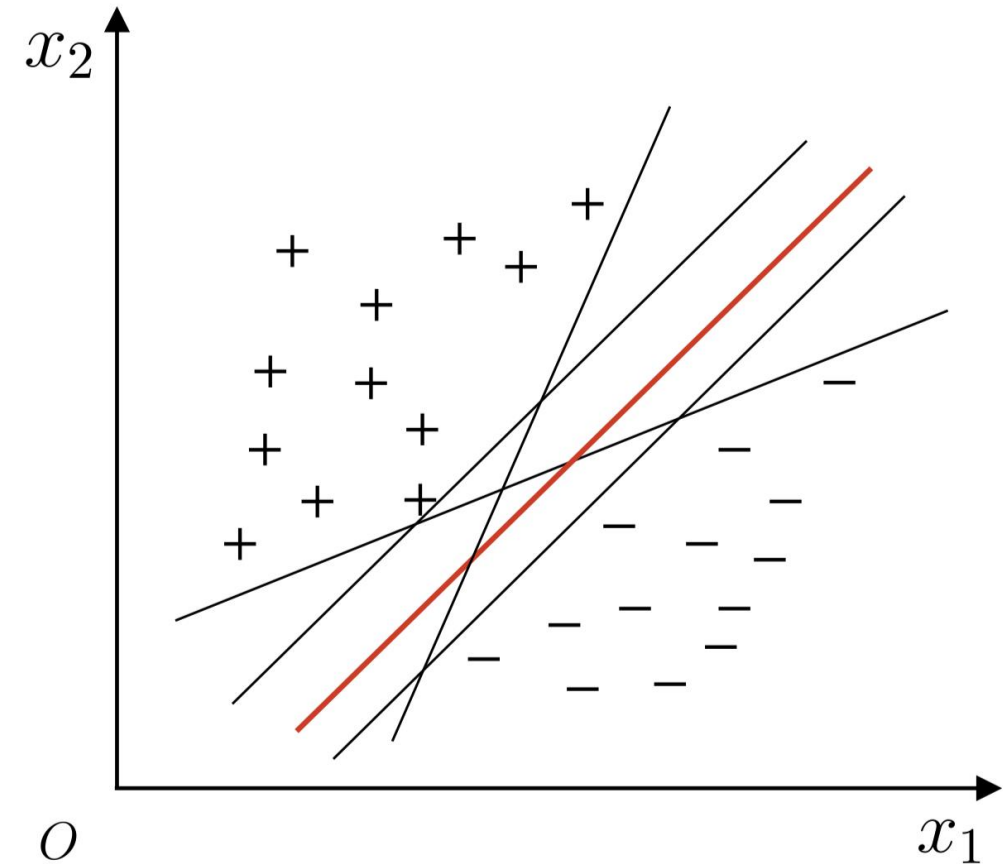
- Move back to binary classification problem
 - Spam email / non-spam email
- Margin
- Support Vector Machines
 - One of the most successful classifiers before deep learning era!
- Warning: While without any proof, this lecture will be very technical. Feel free to interrupt me at any point to ask questions.
- Midterm exam overview

Discussion: which is the best classifier?



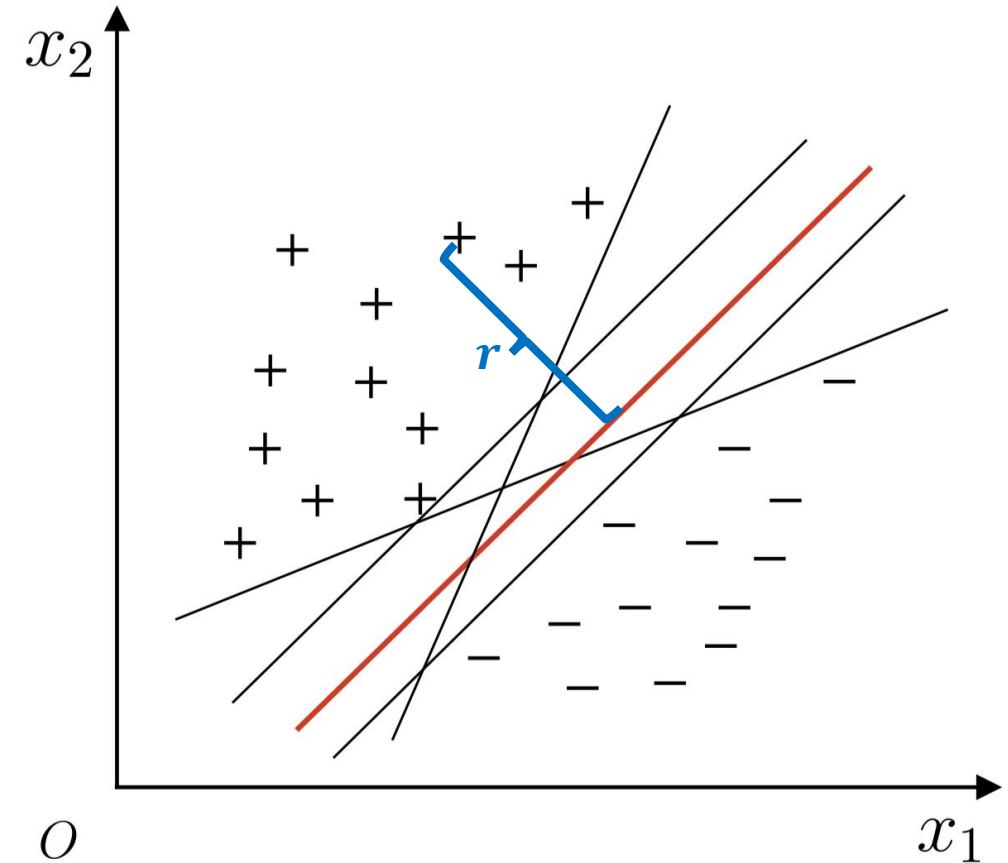
Linear classification

- Input: $x = [x_1, x_2] \in R^2$
- Output: $y \in \{1, -1\}$
- Data: n data points
- Decision line:
 - $w^T x + b = 0$
 - $w \in R^2, b \in R$ are parameters
 - In-class exercise: Rewrite $x_2 = x_1 - 5$ in $w^T x + b = 0$ form.



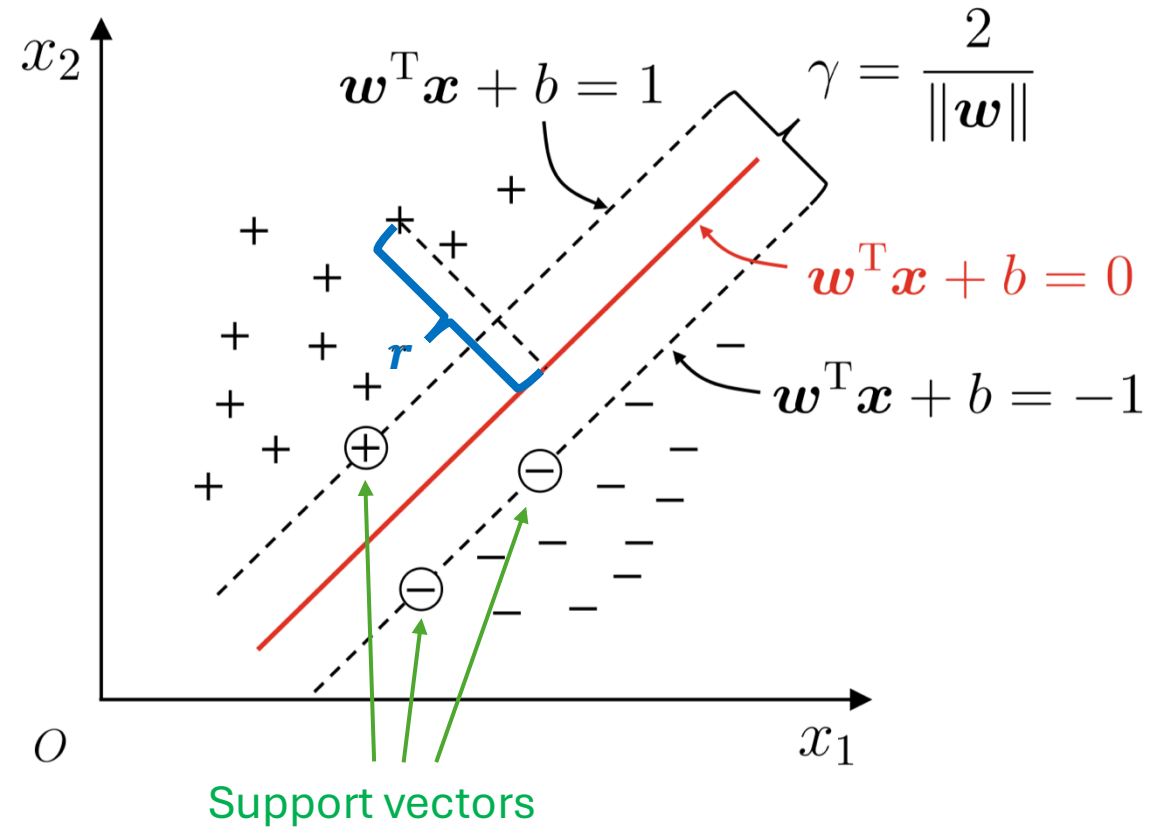
Margin: min distance of data point to line

- Any data point:
 - $x \in R^2$
- Any line:
 - $w^T x + b = 0$
- Margin:
 - $r = \frac{|w^T x + b|}{||w||}$
- **Red** line: We want to learn a max-margin classifier!



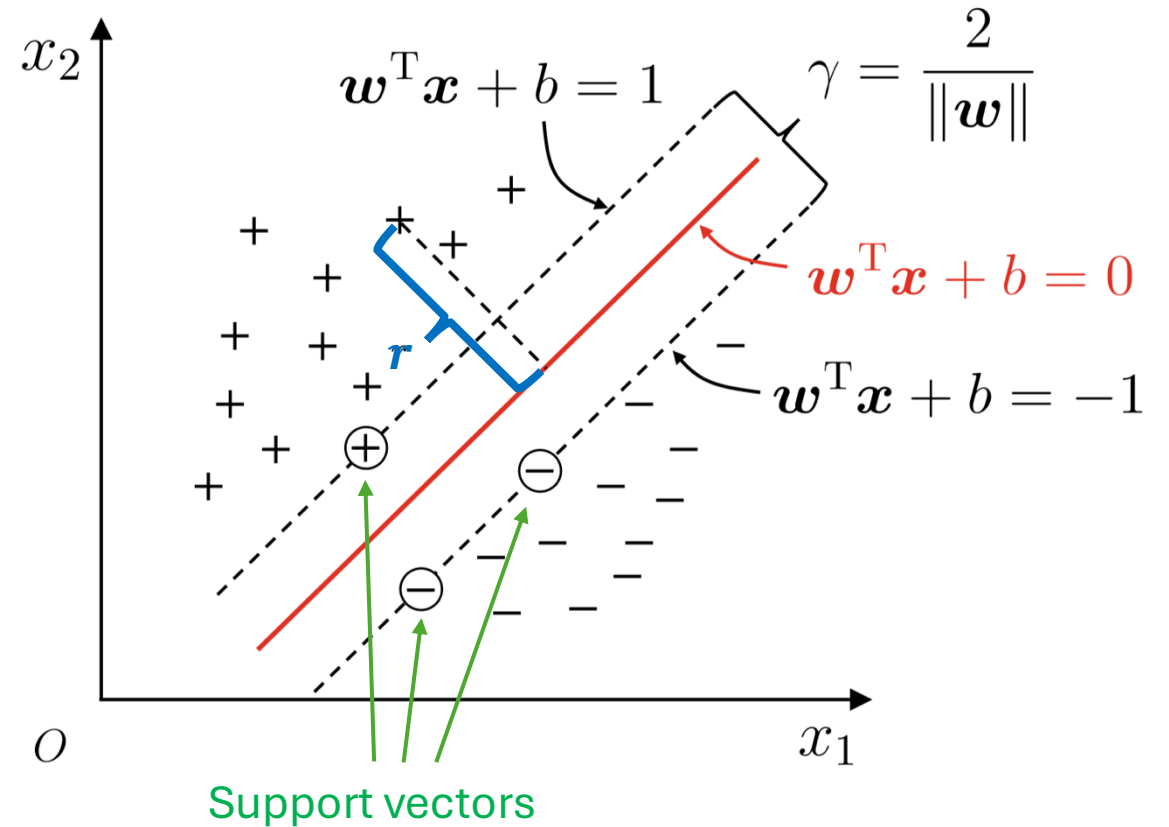
Max-margin classifier

- Discussion: by maximizing margin, which data points are important?
- **Support vectors:**
 - Data points closest to **red** line.
 - Only support vectors affect the training process.
 - Support vector machines (SVM) == max-margin classifier



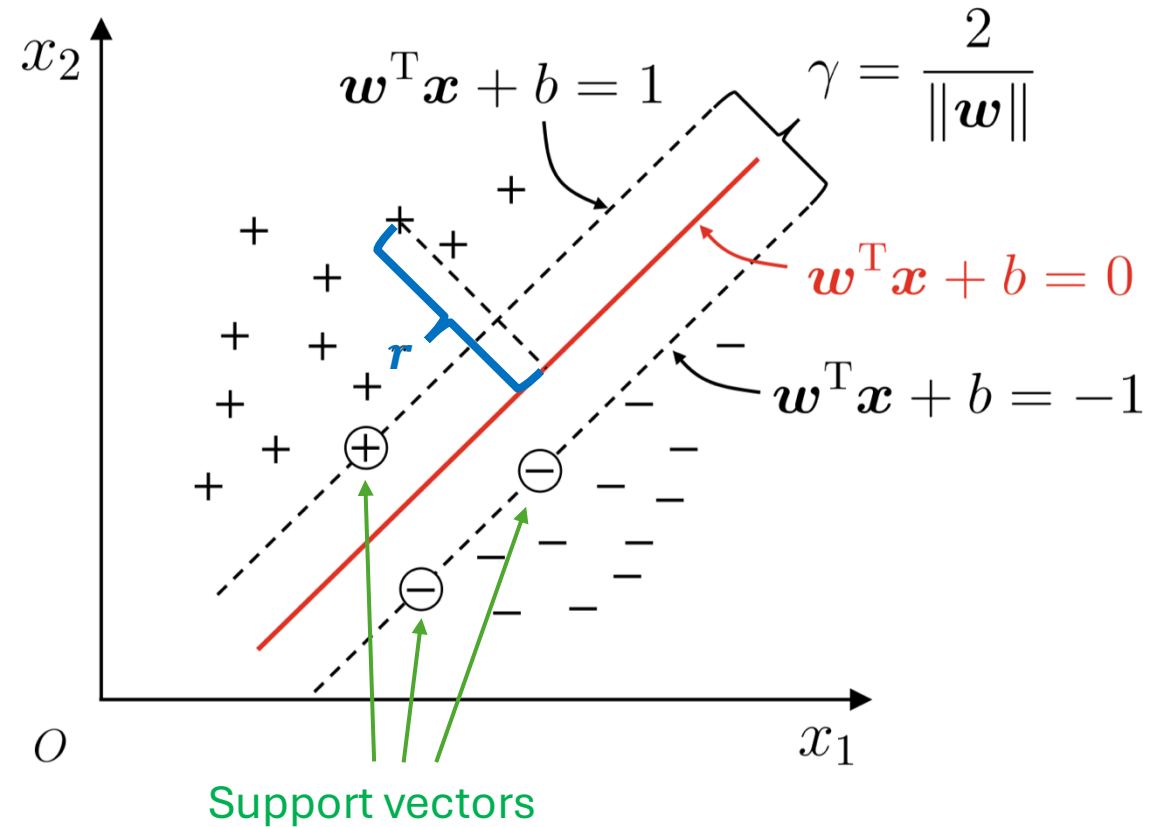
How to train max-margin classifier?

- Assumption:
 - Linearly separable data points
- Recap: Linear classifier
 - If $y = 1, w^T x + b > 0$
 - If $y = -1, w^T x + b < 0$
- Key idea of SVM:
 - If $y = 1, w^T x + b \geq 1$
 - If $y = -1, w^T x + b \leq -1$
 - Why? Support vectors are only data points that matter.



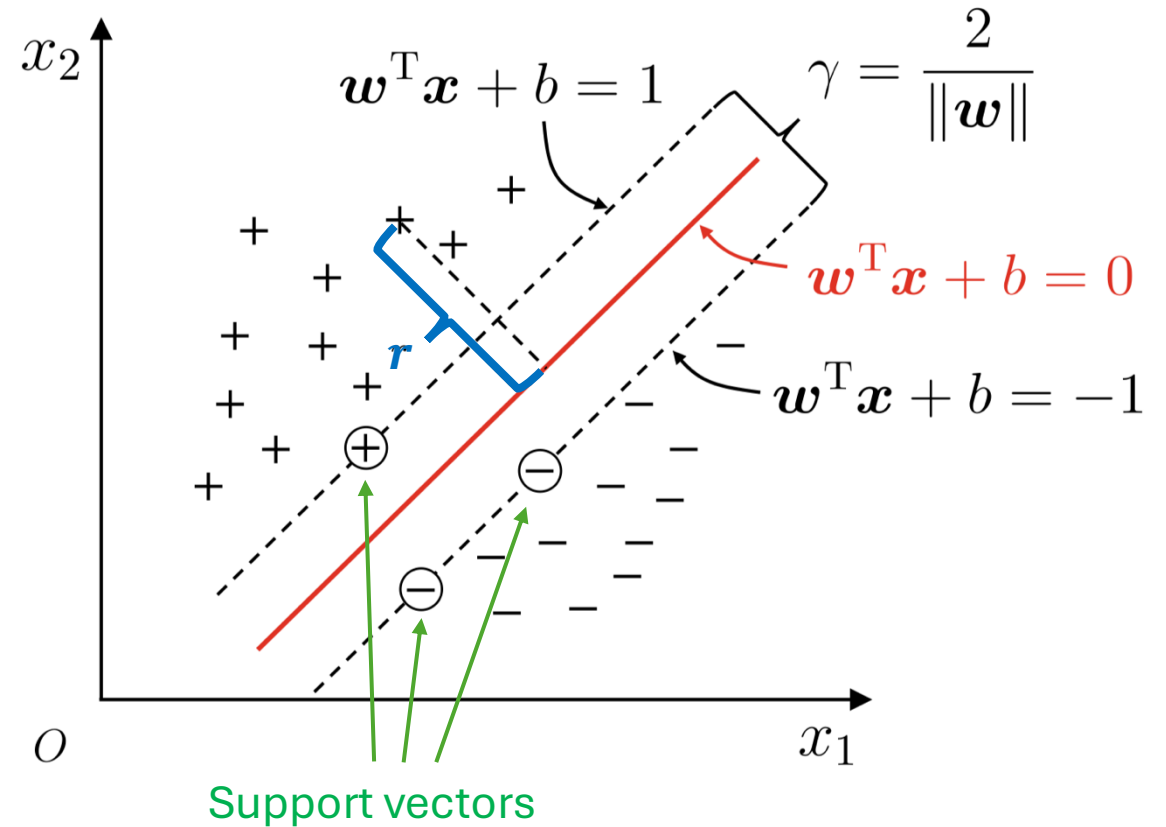
How to train max-margin classifier?

- Key idea of SVM:
 - If $y = 1, w^T x + b \geq 1$
 - If $y = -1, w^T x + b \leq -1$
- Recap: Margin for any data point x
 - $r = \frac{|w^T x + b|}{\|w\|}$
- Total margin between support vectors:
 - $\gamma = \frac{2}{\|w\|}$



How to train max-margin classifier?

- Key idea of SVM:
 - If $y = 1, w^T x + b \geq 1$
 - If $y = -1, w^T x + b \leq -1$
- Total margin between support vectors:
 - $\gamma = \frac{2}{\|w\|}$
- Optimization problem of SVM:
 - $\max_{w,b} \frac{2}{\|w\|}$
 - s. t. $y_i(w^T x_i + b) \geq 1, i = 1, \dots, n$



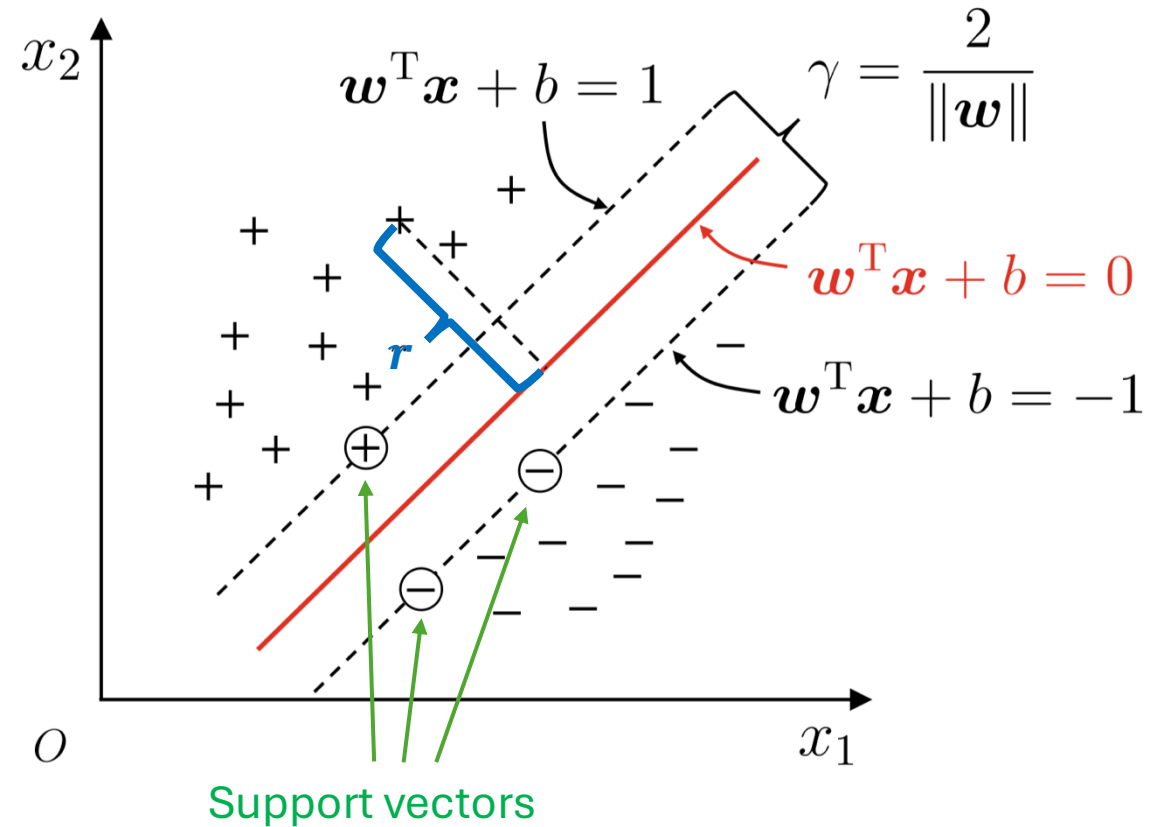
How to train max-margin classifier?

- Optimization problem of SVM:

- $\max_{w,b} \frac{2}{\|w\|}$
- s.t. $y_i(w^T x_i + b) \geq 1, i = 1, \dots, n$

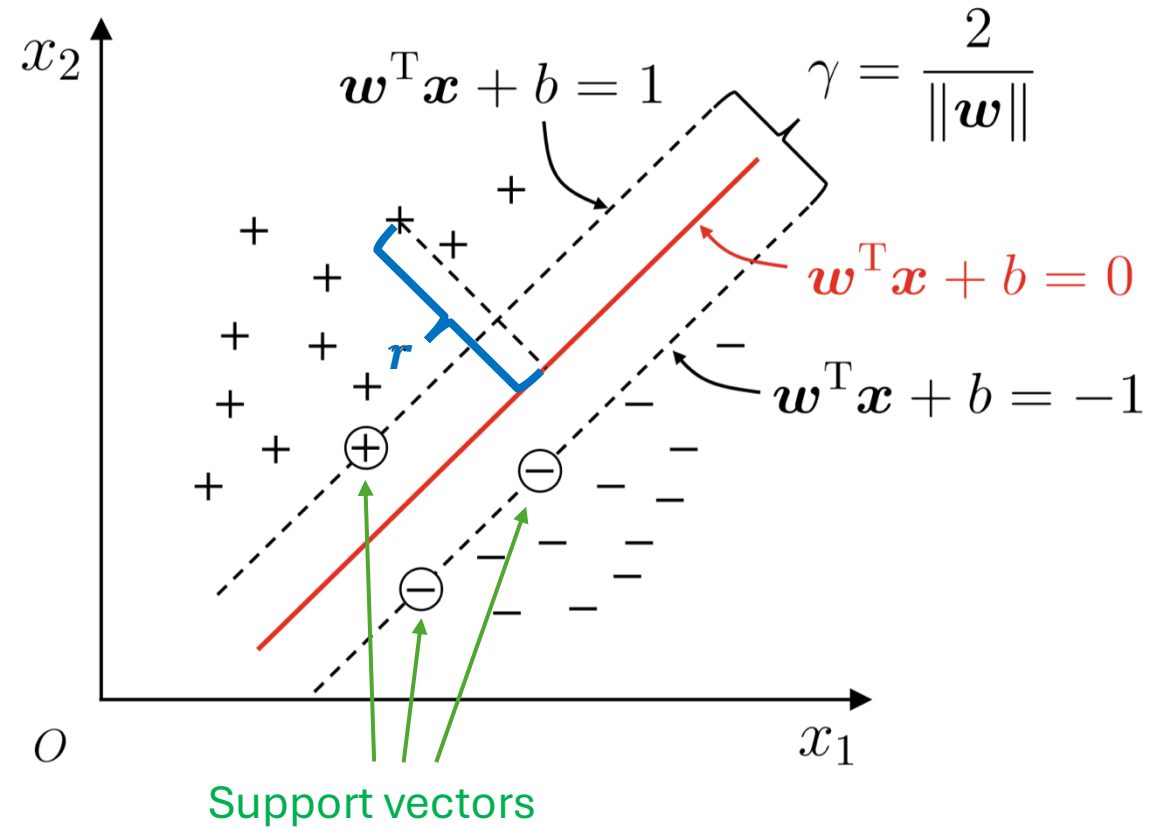
- Equivalent optimization problem:

- $\min_{w,b} \frac{1}{2} \|w\|$
- s.t. $y_i(w^T x_i + b) \geq 1, i = 1, \dots, n$
- Quadratic programming problem
 - Can be solved using some optimization tools, e.g., CPLEX.



How to train max-margin classifier?

- Equivalent optimization problem:
 - $\min_{w,b} \frac{1}{2} \|w\|$
 - s.t. $y_i(w^T x_i + b) \geq 1, i = 1, \dots, n$
- In-class exercise:
 - Write the optimization problem with three support vectors: $(6, 2) +$, $(7, 1) -$, $(8, 2) -$.



Take a deeper look at optimization problem

- Equivalent optimization problem:

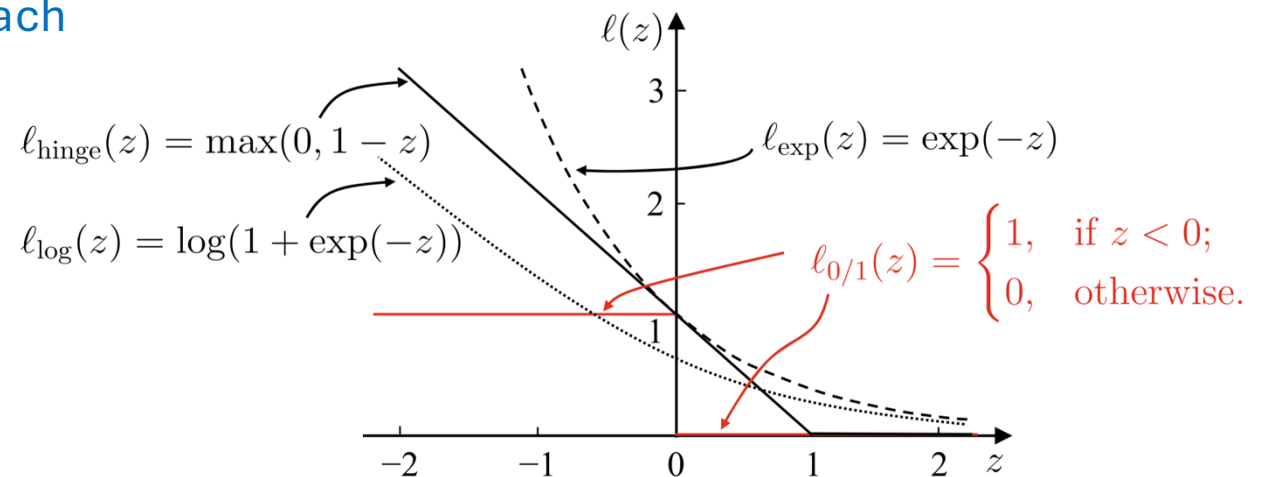
- $\min_{w,b} \frac{1}{2} ||w||$
 - s. t. $y_i(w^T x_i + b) \geq 1, i = 1, \dots, n$

- In-class exercise: Write the Lagrange function

- $\min_{w,b} \frac{1}{2} ||w|| + \sum_{i=1}^n \lambda_i (1 - y_i(w^T x_i + b))$ Related to a New surrogate loss function - Hinge loss!

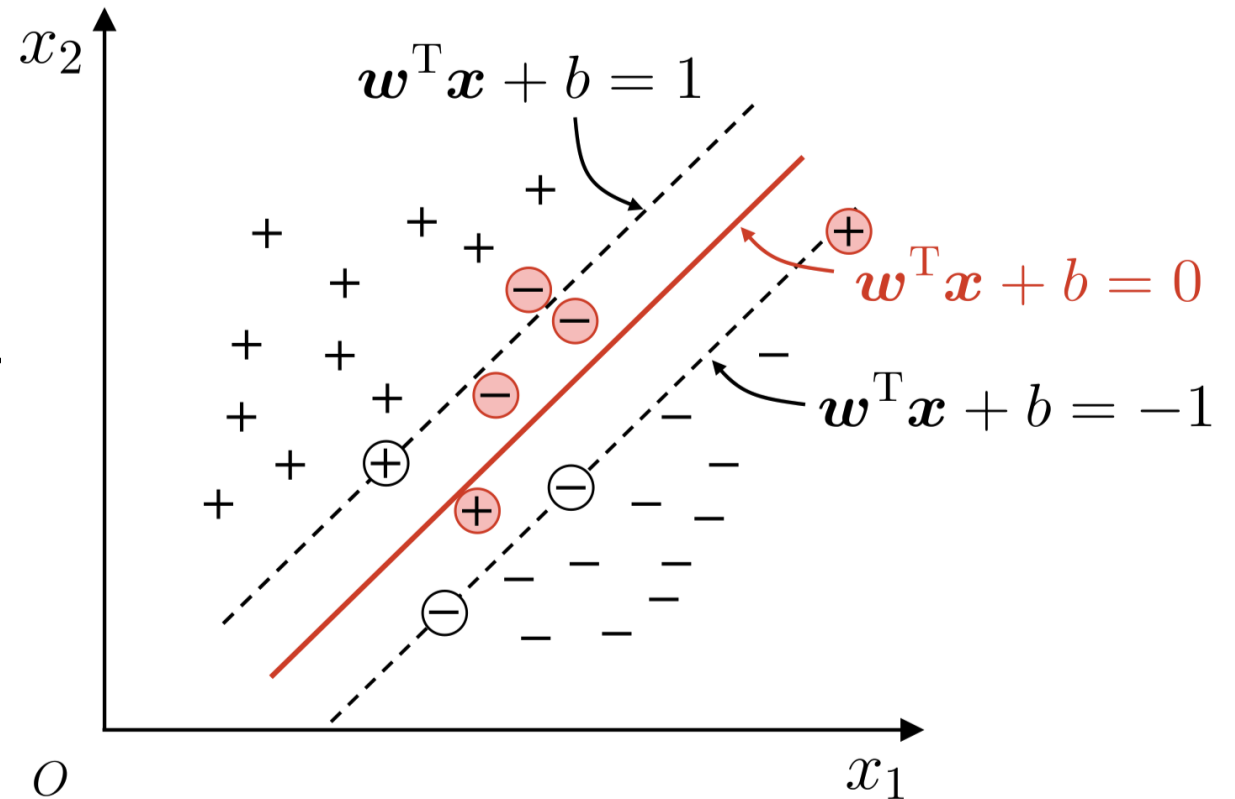
L-2 regularization
on parameter!

Weight of each
data point



What if data points are not linearly separable?

- $y_i(w^T x_i + b) \geq 1$ is violated.
- What can we do?
 - Key idea: we give some tolerance.
- New constraint:
 - $y_i(w^T x_i + b) \geq 1 - \xi$
 - $\xi > 0$
- Discussion: what happens when ξ is very large / small?



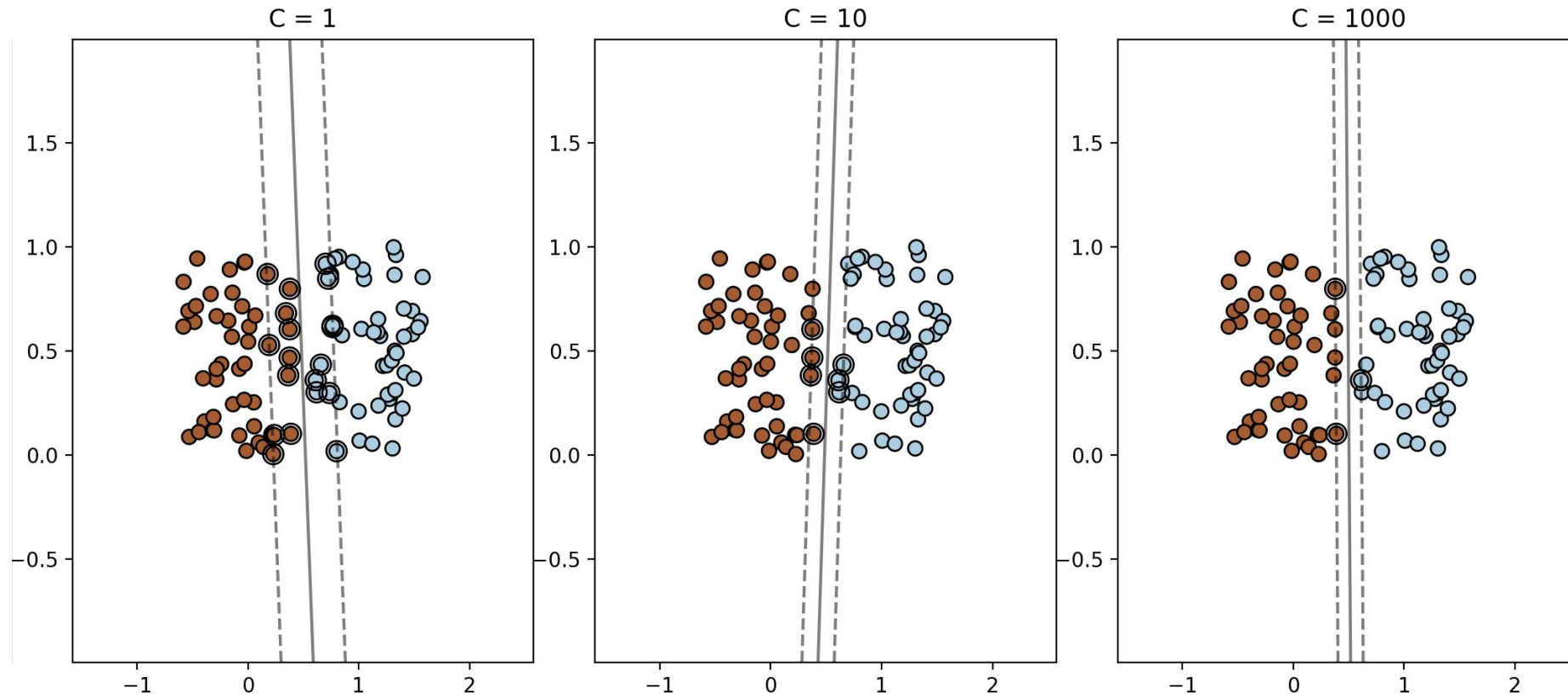
Soft-Margin Support Vector Machines

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i \quad \forall i \in [n] \\ & \xi_i \geq 0 \quad \forall i \in [n] \end{aligned}$$

Equivalent to minimizing **Hinge losses**:

$$\min_{\mathbf{w}, b} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \max [1 - y_i(\mathbf{w}^T \mathbf{x} + b), 0]$$

Hyperparameter C in **soft-margin SVM** and how they affect the margins and “support vectors”.



As C increases, smaller tolerance and fewer soft-margin support vectors.

Checkpoint of Lecture 1-11

- Tasks of ML:
 - Classification (spam / non-spam email) and regression (house price)
- Philosophy of designing ML algorithms:
 - Regularization: Control the complexity of parameters
 - Prevent overfitting
 - Fun fact: L-2 regularization is associated with max margin classifier
 - Optimization: Toolbox of ML
 - ML problem => optimization problem
 - Direct solver, GD, SGD, and much more!
 - Minimize the loss / parameter complexity
 - Maximize the margin

Midterm exam

- What does the exam look like?
 - 80 min (6 - 7:20pm) on Tue Oct 6 here in our classroom
 - Please arrive 5min earlier!
 - Closed-book exam
 - Given individually (not in groups!)
 - Counts 20% towards your final grades
- What to bring?
 - Your pen and calculator (optional).
- What not to bring?
 - Your book, note, lecture slide, or cheat sheet.

What are you expected to know?

- Basic mathematical tools
 - In our math review (Lecture 2-4)
 - Linear algebra, calculus and optimization, probability and statistics

What are you expected to know?

- Basic concepts of machine learning
 - Classification and regression
 - Input space (feature space), output space (label space), hypothesis class
 - Confusion matrix of binary classification
 - Accuracy
 - Holdout / cross validation / hyperparameter
 - Problem of overfitting
 - Loss function
 - Linear model

What are you expected to know?

- Understanding how machine learning algorithms work
 - Why do we need surrogate loss in classification?
 - Why do we need SGD? Drawback of GD?
 - How to define a linear classifier / linear regression?
 - Why do we need SVM? Difference between linear classifier and SVM.
 - Why do we need regularization? How to apply it?
- Important **tips**:
 - Review HW 1 and 2 and all in-class exercise problems.

Midterm course project presentations

- Each group has 5 min
 - 5 min limit is strictly enforced to ensure all groups can present
- Suggested contents:
 - Background, Problem setup, Challenges, Current progress, Plans
- **Evaluation criteria** (each counts 1 point):
 - Soundness, organization, clarity, current progress, and challenges
- Tips for clarity:
 - Figure (most preferred) > tables > bullet points > words > sentences (least preferred)
 - Figure: x-axis and y-axis need to be defined
 - Table: row and column need to be defined